

YEAR 9 SCIENCE
CHEMICAL SCIENCES
REVISION - TEST 2

1. (a) List the properties of an **acid**.

• Produces H^+ ions • Turns blue litmus red.
• Tastes sour •

- (b) List the properties of a **base**.

• Produces OH^- ions. • Soapy feel.
• Tastes bitter. • Turns red litmus blue.

- (c) What is a **salt**?

A substance that comes from an acid reaction.

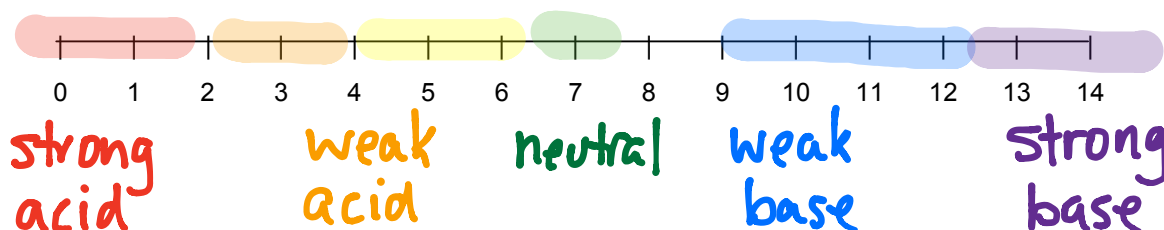
- (d) Classify the following as **ACID**, **BASE** or **SALT**. Use the table below.

ZnCl₂ HCl copper hydroxide H₂SO₄ sodium sulphate calcium acetate
CH₃COOH MgO KOH CuSO₄ HNO₃ aluminium nitrate

ACID	BASE	SALT
HCl CH ₃ COOH H ₂ SO ₄ HNO ₃	copper hydroxide MgO KOH	ZnCl ₂ sodium sulphate calcium acetate CuSO ₄ aluminium nitrate

2. Label the following diagram of the pH scale with the following.

Colour of universal indicator (write in the colours), weak acid, weak base, neutral, strong acid, strong base.



3. Complete the following table, giving **estimates on the pH where necessary**, the **colour of the universal indicator** and whether the substance is a **strong/weak acid**, a **strong/weak base** or **neutral**.

SUBSTANCE	pH	UNIVERSAL INDICATOR (colour)	ACID / BASE/ NEUTRAL (strong / weak)
urine	5-6	yellow	weak acid
pool water	8	light blue	weak base
milk	7	green	neutral
lemonade	2-3	light red	weak acid
drain cleaner	14	purple	strong base
vinegar	3	red	weak acid

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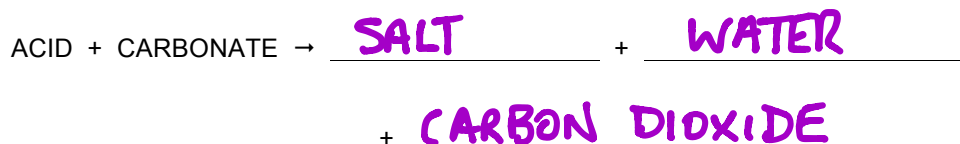
4. (a) Complete the following table for the acids and the salts they produce.

ACID	FORMULA	SALT PRODUCED
hydrochloric	HCl	chloride
nitric	HNO ₃	nitrate
sulphuric	H ₂ SO ₄	sulphate

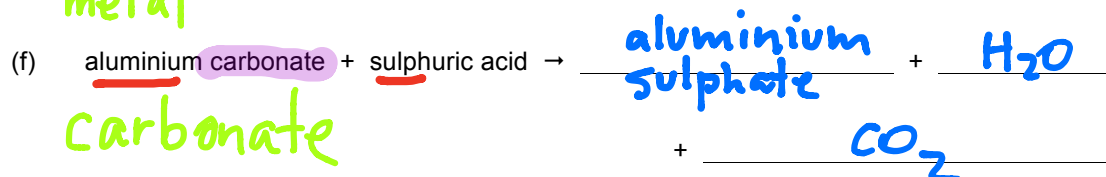
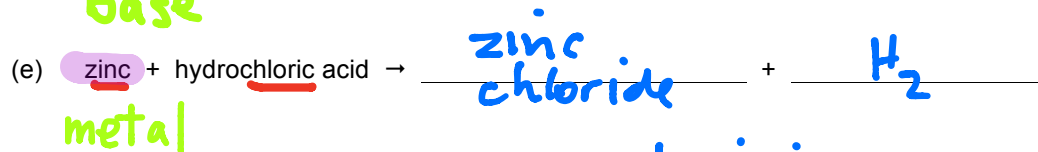
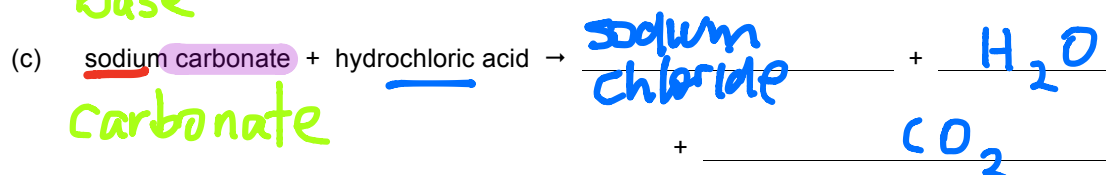
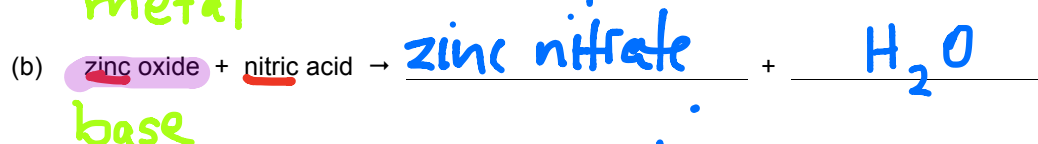
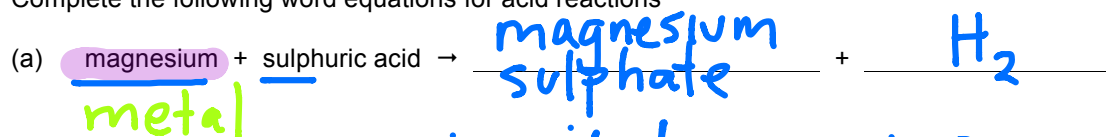
- (b) **Name the salt** produced when the following substances are added to an acid.

REACTION	NAME OF SALT PRODUCED
<u>aluminium</u> hydroxide + <u>sulphuric</u> acid	aluminium sulphate
<u>copper</u> + <u>nitric</u> acid	copper nitrate
<u>calcium</u> hydrogencarbonate + hydro <u>chloric</u> acid	calcium chloride
<u>zinc</u> carbonate + <u>acetic</u> acid	zinc acetate
<u>iron</u> + <u>nitric</u> acid	iron nitrate
<u>magnesium</u> oxide + hydro <u>chloric</u> acid	magnesium chloride

5. Complete the following general word equations for acid reactions.



6. Complete the following word equations for acid reactions



7. **EXTENSION:** Balance the following equations.

